Experiment 7:. Random Sampling and Sampling Distribution

Objective:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

Steps:

1. **Generate a Population:**
    - Create a population of data with a specified distribution (e.g., normal distribution).
2. **Random Sampling:**
    - Perform random sampling from the population to create multiple samples of different sizes.
    - Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. **Sampling Distribution:**
    - Plot histograms or density plots of sample statistics (e.g., sample means).
    - Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. **Central Limit Theorem (Optional):**
    - Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

Example Code:

python

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```
import numpy as np
import matplotlib.pyplot as plt


# Step 1: Generate a population (e.g., normal distribution)
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)


# Step 2: Random sampling
sample_sizes = [30, 50, 100]  # different sample sizes to consider
num_samples = 1000  # number of samples for each sample size
```

```
sample_means = {}

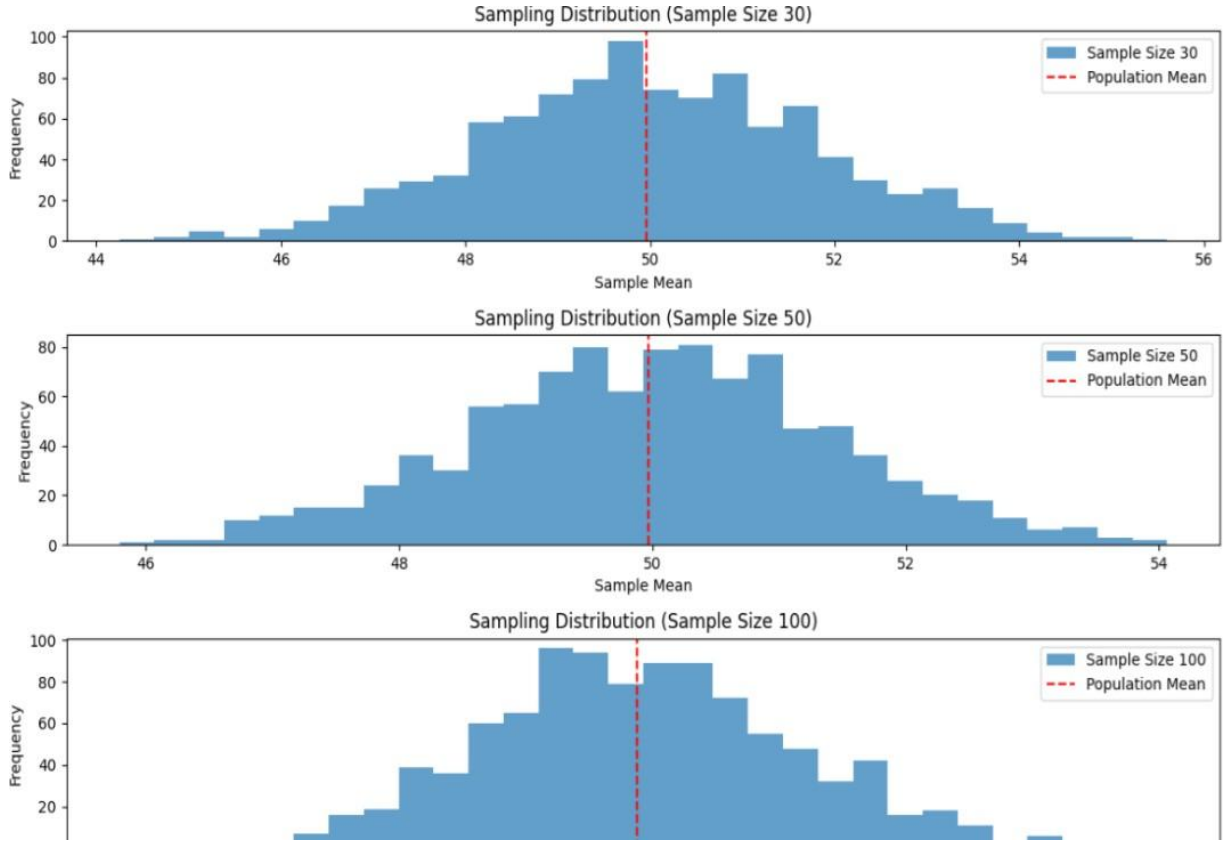
for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i+1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='Population Mean')
    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()
```

OUTPUT:


Sampling Distribution (Sample Size 30)
Frequency
Sample Mean
Sample Size 30
Population Mean
Sampling Distribution (Sample Size 50)
Frequency
Sample Mean
Sample Size 50
Population Mean
Sampling Distribution (Sample Size 100)
Frequency
Sample Size 100
Population Mean